

Customer Behavior & Satisfaction Analysis using Olist E-Commerce Data

1. Introduction:

The Olist Brazilian E-commerce dataset provides detailed information about customer purchases, delivery performance, payments, reviews, sellers, and product categories across Brazil. The objective of this analysis was to understand customer behaviour, identify operational and business patterns, and generate insights that could help improve customer acquisition, retention, and overall customer experience.

To perform this analysis, SQL was used for querying and combining relational tables, Python was used for statistical testing and validation of hypotheses, and Power BI was used to create interactive dashboards for visual storytelling and exploratory analysis.

The analysis focuses on understanding:

- How customers behave across regions ?
- How payment methods influence spending ?
- How delivery experience affects customer satisfaction ?
- How the insights found can support business growth strategies ?

2. Exploratory Data Analysis (EDA) :

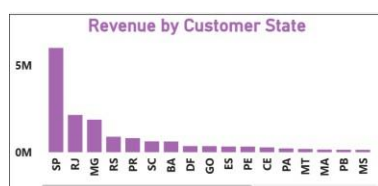
- Overview of Business Performance

The olist dataset contains around 99,000 total orders and customers, generating the total revenue over 16 million. Overall customer satisfaction can be seen as relatively high, with an average review score slightly above 4 out of 5.

The revenue trend shows strong marketplace growth over time, especially after 2017, indicating increasing adoption of the platform and higher transaction activity.

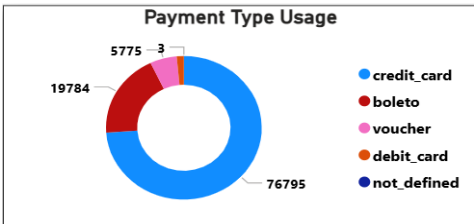


The analysis also shows that customer demand is concentrated in a few major states, with São Paulo contributing the highest share of revenue. This suggests that customer activity and purchasing behavior vary significantly across regions.

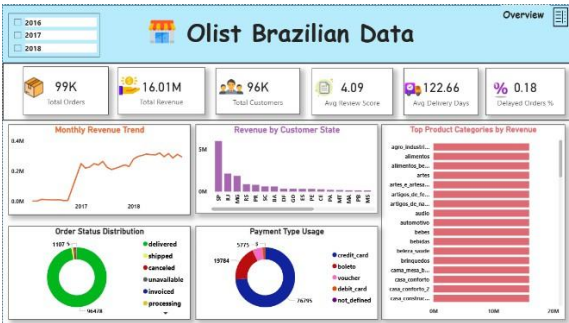


Another important observation from the dashboard is the high amount of credit card payments compared to other payment methods such as boleto, vouchers, and debit cards. This pattern tells us the possible

relationship between payment behavior and transaction value, which was also explored further through hypothesis testing.



Dashboard Overview:



- Customer Experience & Delivery Analysis

Customer reviews show a clear connection with delivery experience. Orders that took longer to reach customers generally received lower review scores, highlighting how important timely delivery is for customer satisfaction.

Delivery performance also varies across different states. Some regions experience longer delivery times and higher freight costs compared to others. Interestingly, freight charges seem to affect customer perception as well, especially in areas where delivery expectations are already high.



Overall, most customers gave positive reviews, with 5-star ratings being the most common. However, even a smaller number of poor reviews can still impact customer trust and seller reputation.

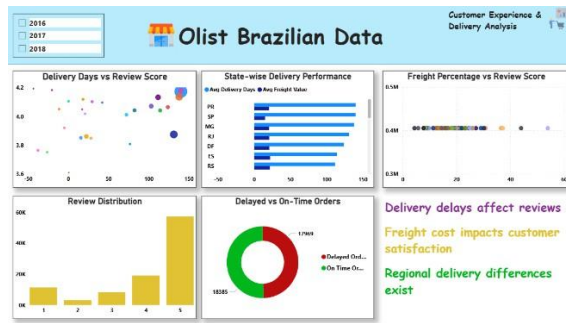


Non-Intuitive Observation

One interesting pattern observed during the analysis was that some regions continued to maintain decent review scores despite having longer delivery times. This suggests that customer satisfaction is not based only on how fast the order arrives, but also on customer expectations with the product.

In some regions, customers may already expect deliveries to take longer and are more accepting of delays. On the other hand, customers in highly urbanized areas often expect faster deliveries and may react more negatively even to smaller delays.

Dashboard Overview:



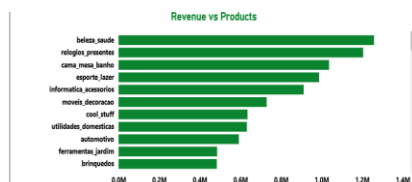
• Seller & Customer Insights

The seller analysis revealed that a small group of sellers contributes a major portion of the overall marketplace revenue. This suggests that the platform depends heavily on its top-performing sellers for consistent business growth.

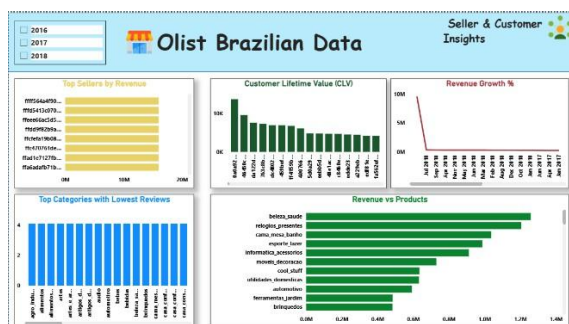
The customer lifetime value analysis also showed that some customers contribute much more revenue than others. This creates an opportunity for the business to identify high-value customers and build more personalized retention and loyalty strategies around them.



Revenue patterns also differ across product categories. Categories such as beauty, home, gifts, and lifestyle products generated comparatively higher revenue, while some categories received lower review scores consistently. This may indicate issues related to product quality, delivery expectations, or overall customer experience in those categories.



Dashboard Overview:



3. Hypothesis Testing :

Hypothesis 1

“Customers Paying through Credit Cards Spend More Than Customers Paying through Other Payment Methods”

• Business Motivation

Different payment methods reflect different customer spending behaviours. Customers using credit cards may feel more comfortable making larger purchases because of benefits like EMI options, cashback offers, and flexible payment facilities.

This analysis was conducted to understand whether payment type has a significant impact on transaction value.

- Null Hypothesis (H_0)

Payment type has no significant effect on transaction value.

- Alternative Hypothesis (H_1)

Customers paying through credit cards have higher average transaction values compared to customers using other payment methods.

- Statistical Test Used

A One-Way ANOVA test was performed to compare average transaction values across different payment methods such as:

- credit card,
- debit card,
- boleto,
- and voucher payments.

The required data was first extracted using SQL and then tested in Python.

F-statistic: 108.28514884260402						
P-value: 5.1823454802529165e-70						
Payment method significantly affects spending behavior						
T-statistic: 19.507526934490898						
P-value: 1.6734956205531664e-84						
OLS Regression Results						

Dep. Variable:	price	R-squared:	0.003			
Model:	OLS	Adj. R-squared:	0.003			
Method:	Least Squares	F-statistic:	108.3			
Date:	Sun, 17 May 2026	Prob (F-statistic):	5.18e-70			
Time:	01:47:27	Log-Likelihood:	-7.7751e+05			
No. Observations:	117329	AIC:	1.555e+06			
Df Residuals:	117325	BIC:	1.555e+06			
Df Model:	3					
Covariance Type:	nonrobust					

	coef	std err	t	P> t	[0.025	0.975]

Intercept	104.1680	1.209	86.161	0.000	101.798	106.538
C(payment_type)[T.credit_card]	22.0845	1.359	16.248	0.000	19.420	24.748
C(payment_type)[T.debit_card]	4.4811	4.603	0.974	0.330	-4.541	13.503
C(payment_type)[T.voucher]	0.1188	2.603	0.046	0.964	-4.983	5.220

Omnibus:	158774.887	Durbin-Watson:	1.805			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	58966181.006			
Skew:	7.662	Prob(JB):	0.000			
Kurtosis:	111.752	Cond. No.	11.1			

Notes:						
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.						

- Result & Interpretation

The statistical results showed that payment methods significantly influence transaction value. Credit card users were found to spend noticeably more compared to customers using other payment methods.

This indicates that flexible payment options and offers on payment methods may encourage customers to make higher-value purchases.

- Business Insight

From a business perspective, this insight helps identifying high-value customer segments. Customers using credit cards may represent stronger revenue opportunities and can be targeted through cashback campaigns, EMI offers, and bank partnerships.

Hypothesis 2

“Customer Satisfaction Is Influenced More by Regional Expectations Than by Delivery Speed Alone”

- Business Motivation

Customer satisfaction is usually expected to depend mainly on delivery speed. However, customer expectations can vary across different regions based on local delivery standards, infrastructure, and buying behaviour.

This analysis was conducted to understand whether customer review scores differ across states even after considering delivery duration and freight costs.

- Null Hypothesis (H_0)

Average customer review scores are the same across all states.

- Alternative Hypothesis (H_1)

At least one state has a significantly different average review score.

- Statistical Test Used

A One-Way ANOVA test was performed using:

- customer state,
- review score,
- delivery duration,
- and freight cost data.

The data was extracted and prepared using SQL, while Python was used for statistical testing.

```
ANOVA RESULTS
F-statistic: 83.98121714275679
P-value: 7.056183174972476e-37
Review scores differ significantly across states

REGRESSION RESULTS
OLS Regression Results
=====
Dep. Variable: review_score R-squared: 0.007
Model: OLS Adj. R-squared: 0.007
Method: Least Squares F-statistic: 43.88
Date: Sun, 17 May 2026 Prob (F-statistic): 1.04e-19
Time: 01:50:44 Log-Likelihood: -20045.
No. observations: 11826 AIC: 4.010e+04
DF Residuals: 11823 BIC: 4.012e+04
DF Model: 2
Covariance Type: nonrobust
=====
coef std err t P>|t| [0.025 0.975]
-----
Intercept 4.3650 0.029 148.860 0.000 4.308 4.423
delivery_days -0.0005 0.000 -2.472 0.001 -0.001 -0.000
freight_value -0.0074 0.001 -8.054 0.000 -0.009 -0.006
=====
Omnibus: 2549.465 Durbin-Watson: 1.593
Prob(Omnibus): 0.000 Jarque-Bera (JB): 4521.200
Skew: -1.459 Prob(JB): 0.00
Kurtosis: 3.815 Cond. No. 458.
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

- Result & Interpretation

The analysis showed noticeable differences in customer satisfaction across regions. The findings suggest that delivery speed alone does not completely explain customer reviews.

In many cases, customer expectations, tolerance toward delays, and regional perception of delivery reliability also appear to influence satisfaction behaviour. This indicates that customer experience may vary depending on regional expectations rather than logistics performance alone.

4. Business Recommendations :

- Promote Credit Card Offers & Cashback Campaigns

The analysis showed that customers using credit cards tend to spend more on average compared to other payment methods. To make the most of this behaviour, the company can collaborate with banks and payment providers to introduce cashback offers, EMI options, and exclusive card-based discounts.

This can help increase average order value while also attracting high-spending customers.

- **Improve Delivery Communication & Transparency**

The findings suggest that customers are more accepting of delays when expectations are communicated clearly. To improve customer experience, the company should focus on providing more accurate delivery estimates, real-time shipment tracking, and proactive delay notifications.

Better communication can reduce customer frustration, improve trust, and create a more consistent delivery experience across regions.

- **Use Region-Specific Customer Experience Strategies**

Customer expectations differ noticeably across states and regions. Instead of following one standard approach nationwide, the company can adopt region-specific communication and logistics strategies based on local delivery expectations and customer behaviour.

This can help improve customer satisfaction and create a more personalized customer experience.

5. Question Requiring Further Exploration

Question: **“Do Credit Card Customers Also Show Higher Long-Term Loyalty and Retention?”**

The analysis showed that credit card users tend to spend more per transaction. However, higher spending does not always mean higher long-term customer value.

This raises an important question:

“Are these customers actually more loyal, or they are simply high one-time spenders?”

The current analysis mainly focuses on payment behaviour and customer satisfaction, but it does not fully explore repeat purchases, retention, churn, or long-term customer lifetime value.

Future analysis could include retention modelling, churn prediction, and customer lifetime value comparison across payment methods. These insights could help the company build stronger lifecycle marketing and customer retention strategies.

6. Appendix :

Tools Used

- SQL Server for relational querying and feature extraction
- Python for statistical testing and analysis
- Power BI for dashboard creation and visualization

LLM Usage Disclosure

ChatGPT was used for:

- Python syntax assistance,
- data-cleaning guidance,
- and visualization formatting suggestions.

All business insights, interpretations, hypotheses, and recommendations were independently developed based on the analysis performed.

